

A Unified Framework for Spatially-Aware Embodied Agents in XR

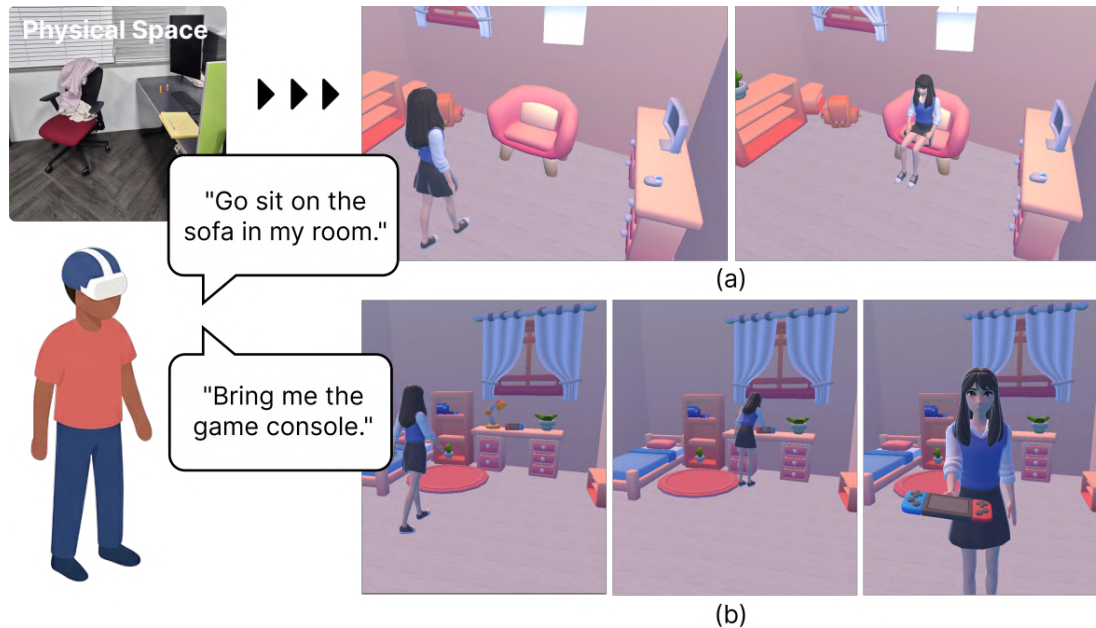
Jihun Shin^{1*}Donghwan Shin^{1†}Eunseong Lee^{1†}Hyeonjin Kim^{1†}Sung-Hee Lee^{1‡}¹KAIST

Figure 1: The figure illustrates Interaction examples with the embodied XR agent. The virtual space (right) is generated based on the user's physical environment (top left). A user wearing an HMD issues voice commands to the agent within this augmented space. (a) The agent responds to "Go sit on the sofa in my room" by navigating to the sofa and sitting down. (b) The agent responds to "Bring me the game console" by walking to the object, picking it up, and delivering it to the user.

ABSTRACT

For embodied AI agents to function effectively in XR environments, they must understand space, navigate within it, and converse and act in contextually appropriate ways. This work presents an XR agent system that integrates these capabilities. The proposed system constructs a virtual environment on top of the user's physical space, enabling the agent to converse with users in natural language and perform appropriate actions while maintaining awareness of spatial context. Through a unified loop where dialogue and motion share the same spatial state, consistency between the agent's utterances and actions is maintained. This work explores how spatially grounded embodied agents can perform natural interaction within personalized XR environments.

Index Terms: Extended Reality, Embodied Agent, Room Augmentation, Human-Agent Interaction

1 INTRODUCTION

Embodied Artificial Intelligence (AI) agents are emerging as a compelling paradigm for interactive systems in spatial computing, par-

ticularly within Extended Reality (XR) [1]. Beyond simple voice or text-based interfaces, agents capable of manifesting as three-dimensional avatars with behavioral context offer new possibilities for human-AI interaction [2]. Central to this vision is spatial grounding—enabling agents to understand the space they occupy, navigate within it, and respond in contextually appropriate ways. This capability constitutes a core technical challenge in the design of embodied AI systems.

A key aspect of spatial grounding is the maintenance of *spatial memory*: a persistent, structured representation of the environment that allows agents to recall object locations, track changes resulting from interactions, and reference shared spatial history in dialogue. As XR technologies enable the capture and playback of immersive spatial experiences, embodied agents that can understand and interact with such memory representations become increasingly important.

Prior work has primarily focused on conversational avatars and static scene understanding through labeling and classification techniques [3, 4]. However, such approaches process spatial information in fixed forms, and systems that integrate dynamically changing spatial context with dialogue history and motion planning remain limited.

We present a prototype system to address these limitations. The proposed system constructs a virtual augmented environment anchored to the user's physical space and implements an agent powered by a Large Language Model (LLM) capable of autonomous navigation within it. To enable spatial understanding, the system encodes scene metadata—spatial layout, object positions, and user

*e-mail: jihun.shin@kaist.ac.kr

†These authors contributed equally to this work.

‡Corresponding author: sunghee.lee@kaist.ac.kr

location—in a structured format and provides this to the LLM to inform language generation. This process is continuously updated in real-time, enabling the agent to dynamically comprehend spatial context. Dialogue and motion modules share this state, allowing the agent to maintain consistency between utterances and actions.

This work explores how AI agents can achieve context awareness and responsiveness within personalized augmented spaces, representing a foundational step toward agents that are not merely conversational but socially present and physically embedded.

2 RELATED WORK

This work builds upon prior research in three key areas: spatial representation and scene graphs, conversational agent systems, and motion planning and generation.

2.1 Spatial Representation and Scene Graphs

Structured spatial representations are essential for agents to understand and reason about 3D environments. Scene graphs, which hierarchically represent object positions, attributes, and inter-object relationships, have served as a core data structure for spatial reasoning.

Recent research has focused on expanding the expressiveness of scene graphs using Vision-Language Models (VLMs) and LLMs. Open3DSG [3] and ConceptGraphs [4] construct open-vocabulary scene graphs from point clouds and RGB-D images, enabling flexible object recognition and relationship inference beyond predefined labels. OpenFunGraph [5] integrates functional relationships (e.g., “the switch controls the light”) into scene graphs beyond spatial relations, while Pix2SG [6] proposes a lightweight approach that extracts semantic relationships directly from images without location information. ESCA [7] presents a framework that leverages scene graphs as intermediate representations to enhance the spatial cognition capabilities of embodied agents.

For dynamic environment tracking, DynaVINS [8] demonstrates methods for tracking changing physical environments through Visual-Inertial SLAM. These works have shown meaningful advances in precise spatial capture and rich semantic representation; however, most are optimized for initial scanning stages or static environment analysis. Scenarios involving continuous updates of real-time spatial states that are immediately reflected in agent interactions remain relatively underexplored.

2.2 Conversational Agents and LLM-based Interaction

Advances in LLMs have significantly expanded the capabilities of conversational agents [1]. Two Tales of Persona in LLMs [9] demonstrates that assigning specific personas to agents enables contextually appropriate response generation, providing a foundation for situated agent design.

Spatial-aware dialogue requires effective methods for conveying environmental information to LLMs. Existing approaches primarily convert object lists or coordinates into text for inclusion in prompts [4, 10]. While this allows LLMs to reference spatial information, additional design considerations are required to transform numerical data into human-centric representations that enable natural deictic expressions.

2.3 Motion Planning and Generation

Translating linguistic intent into physical motion is a core challenge in embodied agent implementation. Language Models as Zero-Shot Planners [10] demonstrates that LLMs can formulate action plans without fine-tuning, while Crafting Dynamic Virtual Activities [2] proposes multi-character behavior generation using multimodal LLMs.

Scene graph-based action planning has also been explored. Long-term Human Trajectory Prediction using 3D Dynamic Scene

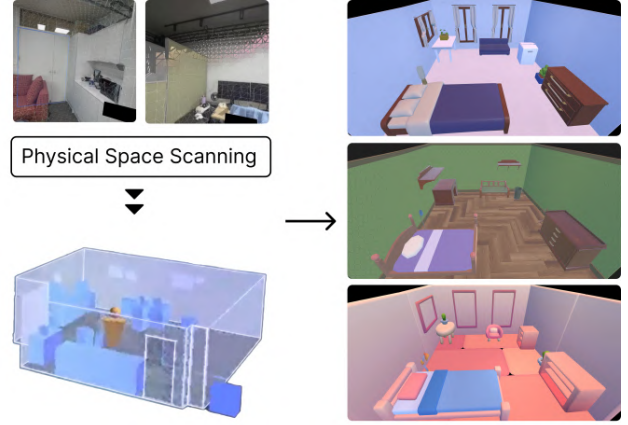


Figure 2: This figure illustrates that in the initialization phase of our framework, spatial information is extracted from the physical real-world space (left), which can then be transformed into various styled spaces sharing the same layout depending on the scenario.

Graphs [11] investigates long-term path prediction leveraging spatial semantics, and X’s Day [12] explores personality-driven daily behavior generation. For scene and context-adapted motion generation, works such as HumanISE [13], Chain-of-Contacts [14], and Move As You Say Interact As You Can [15] present methods for generating interaction motions by combining text commands with 3D environments.

These works have made meaningful progress in language-to-motion translation and scene-aware motion generation. However, most target offline generation or pre-planned scenarios, and integrated systems that simultaneously coordinate speech and action according to real-time dialogue flow and dynamic spatial context remain relatively scarce.

This work presents a system that connects dialogue, motion, and spatial cognition into a unified real-time loop. By providing structured scene descriptions to an LLM-powered agent and sharing the same spatial state between dialogue and motion modules, the system enables spatially grounded multimodal interaction.

3 METHOD

The proposed system consists of two phases: environment initialization and continuous interactive processing (Fig. 4).

3.1 Scene Synthesis

The system utilizes the spatial scanning and Space Setup capabilities of Meta Quest 3 [16] to extract spatial information from the user’s physical environment (Fig. 2). Stylized virtual furniture is synthesized on top of the extracted layout and positioned to correspond with real-world objects. Virtual assets are adjusted in position and scale to match the dimensions of physical furniture.

The system dynamically places assets based on spatial correspondence and maintains a lightweight scene graph $\mathcal{G}$. Scene graphs serve as a data structure for spatial reasoning by hierarchically representing objects, attributes, and relationships [4, 3]. In this system, the scene graph is structured as follows:

$$\mathcal{G} = \{O, P, A\}$$

where $O = \{o_1, o_2, \dots, o_n\}$ denotes the set of objects in the scene, P represents the user state, and A represents the agent state. Each object o_i includes the following attributes: label l_i , world coordinate position $\mathbf{p}_i \in \mathbb{R}^3$, forward direction vector $\mathbf{f}_i \in \mathbb{R}^3$, distances from user and agent d_i^P and d_i^A , and relative directions with

respect to user and agent θ_i^P and θ_i^A . User and agent states each include position $\mathbf{p}$, rotation $\mathbf{q}$, and forward direction vector $\mathbf{f}$.

In XR environments, changes in spatial state originate from the actions of interaction subjects within the space—namely, the user and the agent—in the absence of external physical events. Leveraging this characteristic, the system adopts an update strategy that reflects only transformations arising from interaction events in the scene graph, rather than continuously rescanning the entire environment.

Dynamic manipulation of virtual objects is handled through the game engine’s parent-child hierarchy. Upon execution of a pick command, the target object is re-parented as a child of the agent’s end-effector, synchronizing with the agent’s movement without additional coordinate computation. Upon a place command, the parent-child relationship is released, and the global position and rotation at that moment are recorded in the scene graph. This approach selectively tracks only rigid transformations caused by the actions of interaction subjects, efficiently reflecting dynamic changes following the initial spatial scan.

3.2 Agent Operation

3.2.1 Conversation with Context

The agent’s conversational responses are generated based on contextual information from multiple sources: (1) static agent meta-data, (2) transcribed user speech, (3) recent dialogue history, and (4) real-time state of the scene graph. These inputs are serialized into a structured prompt and transmitted to the LLM backend. The system is implemented in Unity, deployed on Meta Quest 3, and utilizes the OpenAI Realtime API for real-time voice-based dialogue, enabling low-latency speech-to-speech interaction without separate speech recognition and synthesis pipelines.

The approach of assigning a specific persona to an agent to generate contextually appropriate responses has been demonstrated effective in prior work [9].

For effective delivery of spatial context, the system converts numerical coordinates into natural language form. Relative direction is determined by calculating the signed angle between the forward vector of the user or agent and the direction vector toward the object. Specifically, the angle between the forward vector $\mathbf{f}$ and the direction vector $\mathbf{d}_i$ pointing toward the object is computed via `Vector3.SignedAngle`, with the y-component of all vectors flattened to zero to extract only horizontal plane directions. The computed angle is classified into 8-directional sectors at 45° intervals: Forward (-22.5° to 22.5°), Forward-Right (22.5° to 67.5°), Right (67.5° to 112.5°), Back-Right (112.5° to 157.5°), Back ($|\theta| \geq 157.5^\circ$), Back-Left (-157.5° to -112.5°), Left (-112.5° to -67.5°), Forward-Left (-67.5° to -22.5°).

For all objects, both user-relative direction (`playerDirection`) and agent-relative direction (`agentDirection`) are computed simultaneously, providing dual-perspective information. This enables the agent to generate human-centric deictic expressions.

3.2.2 Motion Planning

After dialogue response generation, the system parses the text to extract motion-related intent. Prior work [10] has demonstrated that action planning using LLMs can be performed effectively without fine-tuning. The system extracts an emotional state e and a sequence of action-target pairs (a, t) from the agent’s utterance, structuring them as follows:

$$\tau = \langle e, (a_1, t_1), (a_2, t_2), \dots, (a_k, t_k) \rangle$$

For example, “Follow me” is converted to $\langle \text{smile}, (\text{FOLLOW}, \text{player}) \rangle$, and a compound command such as “pick up the book and sit on the couch” is converted to $\langle \text{smile}, (\text{PICK}, \text{book}), (\text{SIT}, \text{couch}) \rangle$.

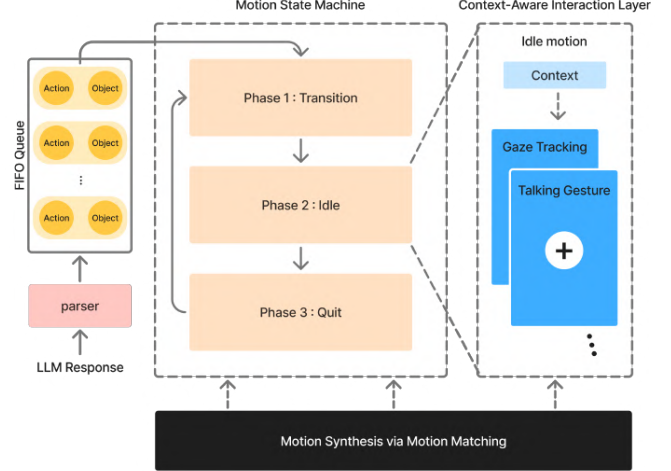


Figure 3: Motion planning pipeline architecture. The diagram shows how LLM responses are parsed into action-object pairs and sequentially processed through a FIFO queue. The motion state machine executes each action via a three-phase cycle (Transition, Idle, Quit), where the Transition phase blocks subsequent actions until completion and the Quit phase executes mandatory exit motions to prepare for the next task. During the Idle phase, a context-aware interaction layer is activated, overlaying gaze tracking toward the user and contextual talking gestures onto the base motion. All state transitions and motions are synthesized via motion matching to ensure seamless animation.

Generated tasks are enqueued in a FIFO(First-In-First-Out) queue and processed sequentially. Agent motion is managed through a motion state machine following a three-phase cycle: Transition, Idle, and Quit (Fig. 3). The Transition phase initiates the action to reach a target (e.g., walking to and sitting on a couch) and functions as a blocking state, preventing subsequent actions until the target pose is fully reached. Once settled, the system enters the Idle phase, maintaining the current posture (e.g., remaining seated while listening). To proceed to the next task in the queue, the system executes the Quit phase, which serves as a mandatory exit step (e.g., standing up from a sitting posture) to terminate the current state and prepare for the next Transition. This cyclic design ensures seamless connectivity between tasks by enforcing necessary exit motions before new actions are initiated.

When dialogue interaction occurs during the Idle phase, a procedural layer is overlaid on the base motion, performing IK-based real-time correction so that the agent’s head, neck, and spine joints orient toward the user’s HMD position, and playing talking flair motion appropriate to the current pose.

Full-body motion is selected through motion matching [17]. Frames most suitable for the current character’s trajectory and pose are retrieved in real-time from the motion database, ensuring natural motion transitions between state changes. Emotional state is reflected in the agent’s facial expressions through blendshape-based control.

3.3 Interactive Loop

The interactive loop supports continuous interaction with the agent. Users speak through voice input without explicit commands, and movement and virtual object manipulation are reflected in the scene graph in real-time. The agent adjusts its gaze toward the user, follows the user when appropriate, and performs contextually relevant actions. This loop supports spatially grounded interaction within the augmented space.

4 APPLICATION AND PERFORMANCE

4.1 Application Scenarios

Because the system constructs the virtual environment based on the user's physical space, the placement of virtual objects spatially corresponds to actual furniture and structures. This allows users to physically navigate the space without colliding with invisible objects, experiencing the virtual environment while maintaining cognitive familiarity with their real surroundings. This physical-virtual spatial alignment serves as a foundation for enhanced immersion.

On this foundation, various application scenarios can be implemented by modifying the visual theme of the virtual environment and the agent's persona settings. The same physical space can be transformed into diverse themes such as a research lab, personal living space, or fantasy environment, while the agent's role and personality are adjusted through persona settings in the prompt. Since alignment with the physical space is maintained, changing only the theme and persona can provide high immersion across different scenarios.

In this work, two scenarios were implemented to verify the system's versatility. The first scenario is a research lab guide assistant. The agent is assigned a junior researcher persona and serves to guide visitors through a shared XR space. The agent provides explanations while referencing the positions of objects within the space and responds to visitor questions in a contextually appropriate manner. The second scenario is a friendly roommate. The agent is assigned a friend persona, engaging in everyday interactions such as introducing each other's spaces and retrieving nearby objects. This scenario demonstrates the potential for social interaction within personalized augmented spaces.

4.2 Performance

The system implements real-time interaction using the OpenAI Realtime API. The entire system operates in real-time at 60 FPS, and response latency was measured for each module.

For voice dialogue, agent responses were generated nearly instantaneously in most situations. Latency of less than 1-2 seconds was observed when utterance content became lengthy; however, this was resolved by applying prompt rules that constrain response length (e.g., maximum 3 sentences).

For motion planning, no perceptible delay occurred during text parsing and action sequence generation. This is a result of processing action intent extraction and queue enqueue operations asynchronously, minimizing the load on the main thread.

5 CONCLUSION

This work presents a prototype system that implements embodied AI agents within personalized, spatially grounded XR environments.

The core contribution of this system lies in connecting dialogue, motion, and spatial cognition into a unified real-time loop. While prior work has focused on individual modalities such as conversational avatars, static scene recognition, and offline motion generation, this work synchronizes these elements within a single system to maintain consistency between the agent's utterances and actions. In particular, the approach of maintaining a lightweight scene graph while selectively tracking only transformations arising from interaction events contributes to computational efficiency in real-time XR environments. Furthermore, by converting numerical coordinates into natural language based on an 8-directional sector system and providing this to the LLM, the agent is enabled to generate human-centric deictic expressions. This design provides versatility, allowing application to various scenarios through modification of persona settings and virtual space styles alone.

From the perspective of XR memory research, this work explores how embodied AI agents can participate in the creation and interaction of spatial memories. The scene graph functions as a shared

Initialization

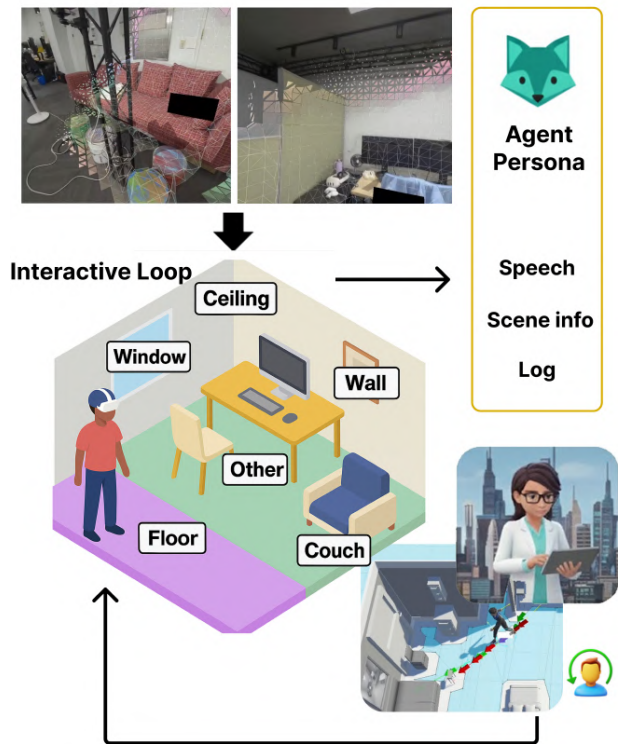


Figure 4: Overall system flowchart illustrating the two-phase architecture. In the initialization phase (top), the physical environment is scanned to extract spatial layout, and an agent persona is configured for the target scenario. In the interactive loop phase (bottom), the scene graph maintains labeled spatial elements (e.g., Floor, Wall, Couch, Window) along with user and agent positions. This spatial information, combined with user speech and dialogue history, is provided to the agent to generate contextually grounded responses and actions. The loop continuously updates as the user moves or manipulates objects, enabling fluid, spatially-aware interaction.

spatial memory representation that both the agent and user can reference, enabling grounded dialogue about “where things are” and “what happened” within the space. As AI agents become integrated into immersive environments, they may serve not only as interaction partners but also as memory aids—recalling object placements, narrating past events, or helping users re-experience recorded spatial contexts. This work represents a step toward understanding the role of embodied agents in capturing, maintaining, and interacting with spatial memories in XR.

However, this work has several limitations. The current system generates motion based on a predefined action set (IDLE, SIT, FOLLOW, PICK, PLACE, etc.), which constrains the range of interactions the agent can perform. Additionally, spatial recognition relies on the static layout from the initial scan, and dynamic changes in the physical environment (e.g., movement of actual furniture) are not reflected. Quantitative evaluation through user studies also remains as future work.

Future research will explore expanding the action set and introducing generative motion synthesis techniques to enhance the agent's expressiveness, as well as investigating spatial recognition methods that reflect dynamic environmental changes in real-time. Additionally, usability evaluations with diverse user groups will be conducted to verify the practical utility of the system.

This work explores the potential of situated AI agents as part of everyday spatial computing experiences, and suggests possibilities for applications in XR-based communication, assistance, and content creation.

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